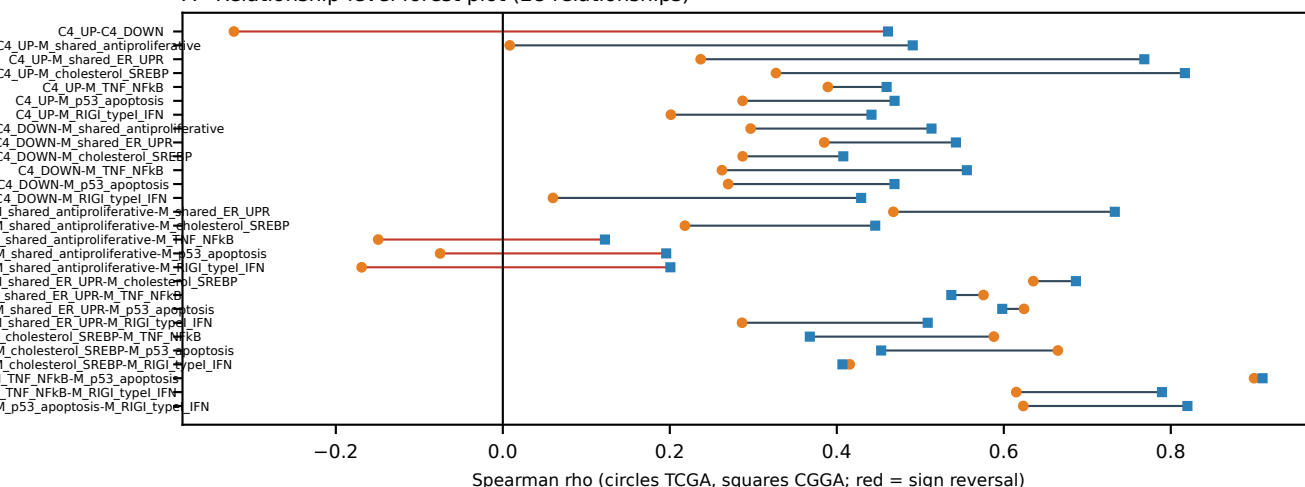
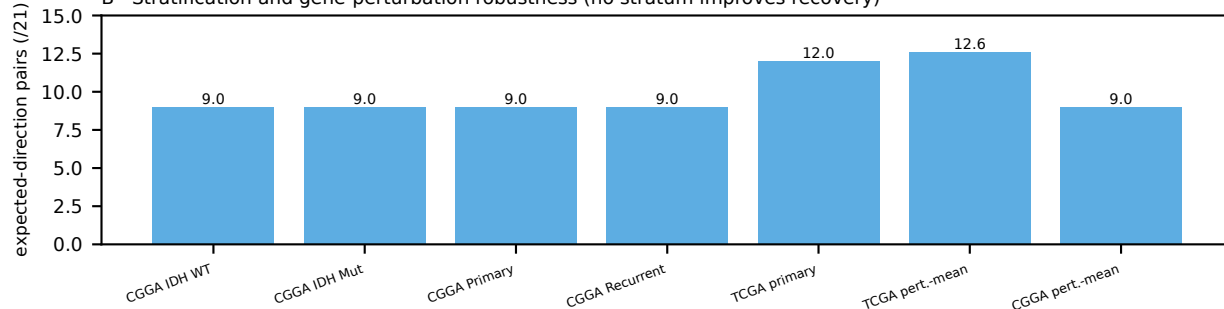


A Relationship-level forest plot (28 relationships)



B Stratification and gene-perturbation robustness (no stratum improves recovery)



C Coverage and partial-subset checks

tcga: min=0.970 (C5_UP) (coverage>=0.80 rule; no imputation; sets unchanged)

cgga: min=0.865 (C5_DOWN) (coverage>=0.80 rule; no imputation; sets unchanged)

Partial (global-mean adjusted) subset: 21/28 TCGA relationships keep their frozen sign

CGGA is treated as contextualisation, not replication; sign-reversed pairs are retained and not relabelled.